



GIRRAWEE HIGH SCHOOL

YEAR 11 - TASK 4

2010

MATHEMATICS

Part A

Time allowed –45 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

Question 1 (10 marks)**Marks**

(a) Evaluate

(i) $\lim_{x \rightarrow -1} (x^2 + 4x)$

1

(ii) $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$

2

(iii) $\lim_{x \rightarrow -4} \left(\frac{x^2 + 2x - 8}{x + 4} \right)$

2

(iv) $\lim_{x \rightarrow \infty} \left(\frac{2x^3 - 3x - 6}{3x^3 + 1} \right)$

2

(b) If $f(x) = x^2 - 4x - 3$, differentiate $f(x)$ from first principles

3

Question 2 (16 marks)

(a) Differentiate

(i) $y = x^3 + 2x^2 - 7x - 3$

1

(ii) $y = x\sqrt{x}$

3

(iii) $y = \frac{x^3 + 2x^2 + 5x - 4}{x^2}$

3

(iv) $y = \sqrt[3]{x^2}$

3

(v) $y = 2x^{\frac{3}{2}} - 5x^{\frac{5}{3}}$

3

(vi) $y = \frac{x-2}{\sqrt{x}}$

3

Question 3 (9 marks)

Differentiate the following:

(i) Use the product rule to differentiate $y = (x^3 + 1)(x^2 - 3x)$ 3

(ii) $y = (2x^3 + 3x^4)^4$ 3

(iii) $y = \frac{x+5}{3x+1}$ 3

Question 4 (10marks)

(i) Find the equations of the *tangent* and *normal* to the curve

$y = x^3 + 2x^2 - 4x - 1$ at the point $(-1, 7)$ 6

(ii) Find the coordinates of the point on the curve $y = 2x(x - 3)$ where the tangent is parallel to the x-axis. 4

Question 5 (8 marks)

(i) Find any point on the curve $y = x^2 + \frac{1}{3}x^3$ where the tangent has the angle of inclination 135° . 3

(ii) The tangent to the curve $x^2 - 3x + 1$, at the point $P(2, -1)$ cuts the y -axis at Q. The normal to the curve at the same point, cuts the y -axis at R. Find the area of $\triangle PQR$ 5



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2010

MATHEMATICS
2 UNIT

Time allowed – 45 minutes

Part B

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new sheet of paper.

Question 1 (10 marks)

Marks

- (a) Find the quadratic equation with roots $2 + \sqrt{3}$ and $2 - \sqrt{3}$. 2
- (b) Write down the equation of a monic quadratic for which one of the zeroes is $x = 1$ and the axis of symmetry is $x = -7$ 2
- (c) Show that the quadratic equation $x^2 + (2k + 3)x + 6k = 0$ has rational roots for all values of k . 3
- (d) For what value(s) of m does the equation $(m - 1)x^2 + 2(m + 1)x + 2m - 1 = 0$ have equal roots? 3

Question 2 (15 marks)

- (a) Solve the equation $3^{2x} + 2 \times 3^x - 15 = 0$ 4
- (b) Find the maximum value of the function $y = 17 + 4x - x^2$ 3
- (c) For what values of k is the expression $kx^2 - (3k - 1)x + k$ positive definite? 3
- (d) For the equation $x^2 + (m - 3)x + m = 0$
- (i) Find an expression for the discriminant $b^2 - 4ac$ in terms of m 1
- (ii) Hence find the values of m for which this equation has:
- (α) exactly one root. 2
- (β) two roots. 2

Question 3.(15 marks)

(a) Find the values of A, B and C if

$$6x^2 - 11 \equiv A(x+2)^2 + Bx + C \quad 4$$

(b) Given that α and β are the roots of the quadratic equation $3x^2 + 4x - 3 = 0$,
Evaluate

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2

(iv) $\alpha^2 + \beta^2$ 2

(v) $\alpha^3\beta + \alpha\beta^3$ 2

(c) Find the value of k in the quadratic equation $(2k-1)x^2 - 5kx + (k+1) = 0$,
if the roots are reciprocal. 3

Year 11 Mathematics 2010

Task 4 Part A solutions

Question 1

$$(i) \lim_{x \rightarrow -1} (x^2 + 4x) = 1 + 4(-1) = -3 \quad (1)$$

$$(ii) \lim_{x \rightarrow 2} \frac{x-2}{x^2-4} = \lim_{x \rightarrow 2} \frac{(x-2)}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{1}{x+2} = \frac{1}{4} \quad (2)$$

$$(iii) \lim_{x \rightarrow -4} \frac{x^2 + 2x - 8}{x+4} = \lim_{x \rightarrow -4} \frac{(x+4)(x-2)}{x+4} = \lim_{x \rightarrow -4} (x-2) = -6 \quad (2)$$

$$(iv) \lim_{x \rightarrow \infty} \frac{2x^3 - 3x - 6}{2x^3 + 1} = \lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x^2} - \frac{6}{x^3}}{2 + \frac{1}{x^3}} = \frac{2}{2} \quad (2)$$

$$(v) f(x) = x^2 - 4x - 3 \\ f(x+h) = (x+h)^2 - 4(x+h) - 3 = x^2 + 2xh + h^2 - 4x - 4h - 3$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - 3 - (x^2 - 4x - 3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 4h}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h - 4)}{h} = 2x - 4 \quad (3)$$

Question 2

$$(i) y = x^3 + 2x^2 - 7x - 3 \\ \frac{dy}{dx} = 3x^2 + 4x - 7 \quad (1)$$

$$(ii) y = x\sqrt{x} = x^{\frac{3}{2}} \\ \frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}} = \frac{3\sqrt{x}}{2} \quad (3)$$

$$(iii) y = \frac{x^3 + 2x^2 + 5x - 4}{x^2} = x + 2 + \frac{5}{x} - \frac{4}{x^2} = x + 2 + 5x^{-1} - 4x^{-2} \\ \frac{dy}{dx} = 1 - 5x^{-2} + 8x^{-3} = 1 - \frac{5}{x^2} + \frac{8}{x^3} \quad (3)$$

$$(iv) y = \sqrt[3]{x^2} = x^{\frac{2}{3}} \\ \frac{dy}{dx} = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}} \quad (3)$$

$$(v) y = 2x^{\frac{3}{2}} - 5x^{\frac{5}{3}} \\ \frac{dy}{dx} = 2 \cdot \frac{3}{2} x^{\frac{1}{2}} - 5 \cdot \frac{5}{3} x^{\frac{2}{3}} = 3x^{\frac{1}{2}} - \frac{25}{3} x^{\frac{2}{3}} \quad (3)$$

Question 2

$$(vi) \quad y = \frac{x-2}{\sqrt{x}} = \sqrt{x} - \frac{2}{\sqrt{x}}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2}x^{-\frac{1}{2}} - 2 \cdot \left(-\frac{1}{2}\right)x^{-\frac{3}{2}} \\ &= \frac{1}{2\sqrt{x}} + \frac{1}{x^{\frac{3}{2}}} = \frac{1}{2\sqrt{x}} + \frac{1}{\sqrt{x^3}} \end{aligned}$$

(3)

Question 3

$$(i) \quad y = (x^3+1)(x^2-3x)$$

$$\begin{aligned} \frac{dy}{dx} &= vu' + uv' \quad \begin{array}{l} v = x^3+1 \\ v' = 3x^2 \\ u = x^2-3x \end{array} \\ &= (x^3+1)(2x-3) + (x^2-3x)3x^2 \quad u' = 2x-3 \end{aligned}$$

$$\begin{aligned} &= 2x^4 + 2x - 3x^3 - 3 + 3x^4 - 9x^3 \\ &= 5x^4 - 12x^3 + 2x - 3 \end{aligned}$$

(3)

$$(ii) \quad y = (2x^3+3x^4)^4$$

$$\begin{aligned} \frac{dy}{dx} &= 4 \cdot (2x^3+3x^4) \cdot (6x^2+12x^3) \\ &= \cancel{4x^3} \cdot \cancel{(2x^3+3x^4)} \cdot 6x^2(1+2x) \\ &= 24x^3(2+3x) \cdot 6x^2(1+2x) \\ &= 24x^5(2+3x)(1+2x) \end{aligned}$$

(3)

$$(iii) \quad y = 4x(3x-2)^5$$

$$v = 4x$$

$$v' = 4$$

$$u = (3x-2)^5$$

$$u' = 15(3x-2)^4$$

$$\begin{aligned} \frac{dy}{dx} &= vu' + uv' \\ &= 4x \cdot 15(3x-2)^4 + (3x-2)^5 \cdot 4 \end{aligned}$$

$$= 60x(3x-2)^4 + 4(3x-2)^5$$

$$= 4(3x-2)^4(15x+3x-2)$$

$$= 4(3x-2)^4(18x-2)$$

$$= 8(3x-2)^4(9x-1) \quad (3)$$

$$(iv) \quad y = \frac{x+5}{3x+1}$$

$$u = x+5$$

$$u' = 1$$

$$v = 3x+1$$

$$v' = 3$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{(3x+1) \cdot 1 - (x+5) \cdot 3}{(3x+1)^2}$$

$$= \frac{3x+1-3x-15}{(3x+1)^2} = \frac{-14}{(3x+1)^2} \quad (3)$$

Question 4

(i) $y = x^3 + 2x^2 - 4x - 1$

$$\frac{dy}{dx} = 3x^2 + 4x - 4$$

at $x = -1$, gradient $m =$
 $= 3(-1)^2 + 4(-1) - 4$
 $= 3 - 4 - 4 = -5$

equation of tangent Pt $(-1, 7)$

$$y - 7 = -5(x + 1) = -5x - 5$$

ie $5x + y - 2 = 0$ (3)

gradient of normal $= \frac{1}{5}$

equation of normal

$$y - 7 = \frac{1}{5}(x + 1)$$

$$5y - 35 = x + 1$$

ie $x - 5y + 36 = 0$ (3)

(ii) $y = 2x(x - 3) = 2x^2 - 6x$ (1)

$$\frac{dy}{dx} = 4x - 6$$

Parallel to x -axis $\Rightarrow$ gradient $= 0$

$$\therefore 4x - 6 = 0 \Rightarrow x = \frac{3}{2}$$

sub $x = \frac{3}{2}$ in (1)

$$y = 2 \cdot \frac{3}{2} \left(\frac{3}{2} - 3 \right) = 3 \cdot \left(-\frac{3}{2} \right)$$

$$= -\frac{9}{2}$$

Co-ordinates are $\left(\frac{3}{2}, -\frac{9}{2} \right)$.

Question 5

(i)

$$y = x^2 + \frac{1}{2}x^3$$

$$\frac{dy}{dx} = 2x + \frac{1}{2}x^2$$

gradient $= \tan 135^\circ = -1$

$$\therefore 2x + \frac{1}{2}x^2 = -1$$

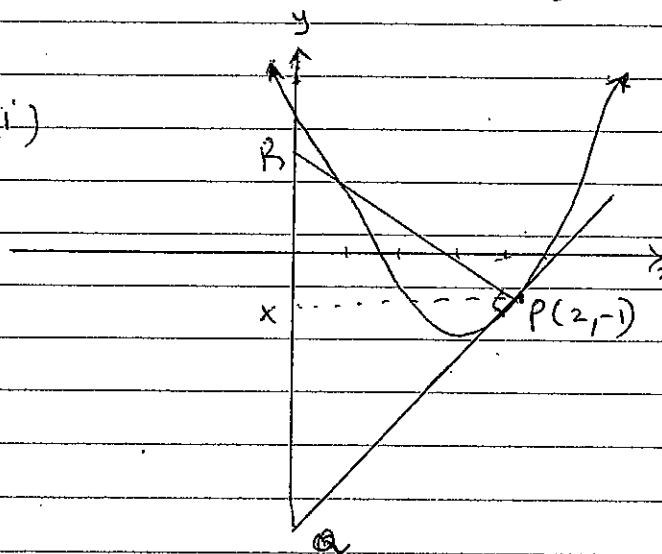
$$x^2 + 2x + 1 = 0 \Rightarrow (x + 1)^2 = 0$$

$$\therefore x = -1$$

$$\therefore y = (-1)^2 + \frac{1}{2}(-1)^3 = 1 - \frac{1}{2} = \frac{1}{2}$$

$\therefore$ the point is $\left(-1, \frac{1}{2} \right)$

(ii)



$$y = x^2 - 3x + 1 \quad P = (2, -1)$$

$$\frac{dy}{dx} = 2x - 3$$

at P , gradient $= 4 - 3 = 1$

equation of tangent

$$y + 1 = 1(x - 2) \Rightarrow y = x - 3$$

$\therefore$ co-ordinates of $Q = (0, -3)$.

eqn of normal at P is

$$y + 1 = -1(x - 2) \Rightarrow y = -x + 1$$

co-ordinates of $R = (0, 1)$.

Area of $\Delta PQR = \text{area}(\Delta PRx) + \text{area}(\Delta PQx)$
 $= 4u^2 = \frac{1}{2} \cdot 2 \cdot 2 + \frac{1}{2} \cdot 2 \cdot 2$

Year 11 mathematics.

Task 4 2010 solutions.

Part B

Question 1.

a) roots: $2+\sqrt{3}$, $2-\sqrt{3}$

$$\alpha + \beta = 4 ; \alpha\beta = 4 - 3 = 1$$

$$\therefore \text{equation: } x^2 - 4x + 1 = 0$$

(2)

b) $y = (x+7)^2 + k$

one zero is at $x=1$.

sub in $x=1$

$$\therefore 0 = 8^2 + k \Rightarrow k = -64$$

$$\therefore y = (x+7)^2 - 64$$

$$y = x^2 + 14x - 15$$

(2)

c) $x^2 + (2k+3)x + 6k = 0$

$$\Delta = (2k+3)^2 - 4 \cdot 1 \cdot 6k$$

$$= 4k^2 + 12k + 9 - 24k$$

$$= 4k^2 - 12k + 9$$

$$= (2k-3)^2 \text{ is a perfect}$$

square,

$\therefore$ has rational roots for all k

d) $(m-1)x^2 + 2(m+1)x + 2m-1 = 0$

$$\text{equal roots} \Rightarrow b^2 - 4ac = 0$$

$$\therefore 4(m+1)^2 - 4(m-1)(2m-1) = 0$$

$$\Rightarrow 4[m^2 + 2m + 1 - 2m^2 + 3m - 1] = 0$$

$$\Rightarrow 4[-m^2 + 5m] = 0, \dots \dots \dots$$

Question 2

a) $\frac{2x}{3} + 2 \times 3^x - 15 = 0$

$$\text{Let } u = 3^x$$

$$\text{then } u^2 + 2u - 15 = 0$$

$$(u+5)(u-3) = 0$$

$$u = -5, 3$$

$$\therefore 3^x = 3 \text{ or } 3^x = -5$$

$$3^x = 3^1 \text{ or no solution}$$

$$\therefore x = 1$$

(4)

b) $y = 17 + 4x - x^2$

$$= -(x^2 - 4x - 17)$$

$$= -[(x-2)^2 - 4 - 17]$$

$$= -(x-2)^2 + 21$$

$$\therefore \text{maximum value} = 21$$

(3)

c) $15x^2 - (3k-1)x + k$

positive definite $\Rightarrow a > 0, \Delta < 0$

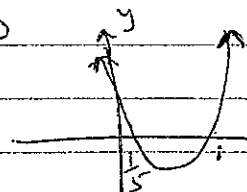
$$\therefore k > 0 \text{ and } \Delta = (3k-1)^2 - 4k^2 < 0$$

$$9k^2 - 6k + 1 - 4k^2 < 0$$

$$5k^2 - 6k + 1 < 0$$

$$(5k-1)(k-1) < 0$$

$$\frac{1}{5} < k < 1$$



(3)

Question 2

d) $x^2 + (m-3)x + m = 0$

(i) $D = b^2 - 4ac = (m-3)^2 - 4m$
 $= m^2 - 10m + 9$ ①

(ii) (a) exactly one root $\Rightarrow D = 0$

$$\therefore m^2 - 10m + 9 = 0$$

$$(m-9)(m-1) = 0$$

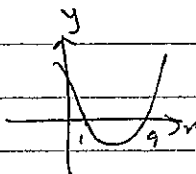
$$\therefore m = 1 \text{ or } 9$$

(b) two roots $\Rightarrow D > 0$

$$m^2 - 10m + 9 > 0$$

$$(m-9)(m-1) > 0$$

$$\therefore m > 9 \text{ or } m < 1$$



Question 3

a) $6x^2 - 11 = A(x+2)^2 + Bx + C$
 $= A(x^2 + 4x + 4) + Bx + C$
 $= Ax^2 + x(4A+B) + 4A+C$

Equating the coefficients of same powers of x ,

$$A = 6; 4A + B = 0 \Rightarrow B = -24$$

$$4A + C = -11 \Rightarrow C = -35$$

$$\therefore A = 6; B = -24; C = -35$$

④

b) $3x^2 + 4x - 3 = 0$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{4}{3}$ ①

(ii) $\alpha\beta = \frac{c}{a} = \frac{-3}{3} = -1$ ①

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-\frac{4}{3}}{-1} = \frac{4}{3}$ ②

(iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= \frac{16}{9} + 2 = \frac{34}{9} = 3\frac{7}{9}$ ②

(v) $\alpha^3\beta + \alpha\beta^3 = \alpha\beta(\alpha^2 + \beta^2)$
 $= -1\left(\frac{34}{9}\right) = -\frac{34}{9}$
 $= -3\frac{7}{9}$ ②

c) $(2k-1)x^2 - 5kx + k+1 = 0$

Let the roots be α and $\frac{1}{\alpha}$

$$\alpha\beta = \frac{c}{a}$$

$$\therefore \alpha \times \frac{1}{\alpha} = 1 = \frac{k+1}{2k-1}$$

$$\therefore 2k-1 = k+1$$

$$\therefore k = 2$$
 ③